

README COMPUTATIONAL NEUROSCIENCE

Martina Napoli

Perceptual decisions are usually based on current sensory information, but previous choices may also influence later judgments. This project examines whether repeated perceptual decisions depend only on new evidence or are biased by previous choices. Participants performed a visual task in which they classified sequences of oriented stimuli as cardinal or diagonal across repeated presentations. The design allowed us to test whether decision stability reflected simple response repetition or a biased integration of new sensory evidence.

Behavioral results showed that participants often repeated previous decisions, even when the stimulus order changed. However, they remained sensitive to the current sensory evidence, suggesting that later choices combined new information with previous decisions. Confidence also contributed to this effect, as high-confidence decisions were more likely to be repeated. Computational modeling showed that previous decisions were best explained as influencing evidence accumulation rather than only the initial response bias. Machine learning and EEG analyses provided complementary evidence that decision history and neural activity contained information about later behavior.

Overall, these findings suggest that perceptual confirmation bias stabilizes previous interpretations and influences how new sensory evidence is used.

Dataset

- CJ_EEG.csv → raw dataset: 6600 rows x 36 columns
- CJ_EEG_clean.csv → cleaned dataset after RT outlier removal using mean + 4 SD per participant

Key columns :

Column	Description
subj	Subject ID. The dataset includes 25 participants, although subject labels range from s01 to s27 because some subject numbers were not included in the final sample.
nrep	Presentation number: 0 = first, 1 = second, 2 = third.
cond	True stimulus condition: 0 = cardinal, 1 = diagonal.
deci	Participant decision: 0 = cardinal, 1 = diagonal.
rDV	Raw decision variable representing sensory evidence strength.
d1-d6	Evidence samples per presentation.
confi	Confidence rating.
RT	Reaction time in seconds.
metad	Meta-d prime, reflecting metacognitive sensitivity per subject.
deci-1, deci-2, deci-3	Previous decisions used to represent decision history.
corr-1	Correctness of the previous decision.

Repository structure

```
Computational Neuroscience Martina Napoli/  
|  
|-- DATA/  
|   |-- CJ_EEG.csv  
|   |-- CJ_EEG_clean.csv  
|   |-- preprocessament/EEG/  
|   |   |-- s01/ - s27/  
|   |   |   |-- sXX_preproc.ipynb  
|   |   |   |-- sXX_bids.ipynb  
|   |   |   |-- sXX_eye.ipynb  
|   |   |-- main_epo.fif          # Main task epochs  
|   |   |-- mainstim_epo.fif     # Stimulus-locked epochs
```

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| | | |-- mainresp_epo.fif          # Response-locked epochs
| | | |-- loc_epo.fif              # Localiser epochs
| | | |-- locstim_epo.fif          # Localiser stimulus-locked epochs
| |-- decoding_results_*.pkl
|
|-- BEHAVIOURAL ANALYSES/
| |-- behaviour_analyses_TFG_MARTINA.ipynb
| |-- behaviour_results/
| | |-- summary_behavioural.csv
| | |-- plot_data_panelA.csv
| | |-- plot_data_panelB.csv
| | |-- plot_data_panelC.csv
|
|-- EEG ANALYSES/
| |-- EEG_initial.ipynb
| |-- EEG_analysis_TFG_MARTINA.ipynb
| |-- EEG_key_analyses_TFG_MARTINA.ipynb
| |-- EEG_decoding_decision_TFG_MARTINA.ipynb
| |-- EEG_decoding_DL_TFG_MARTINA.ipynb
| |-- eeg_results/
| | |-- eeg_auc_main_per_subject.csv
| | |-- eeg_auc_per_subject.csv
| | |-- eeg_decoding_temporal.csv
|
|-- ML ANALYSES/
| |-- ML_analyses_TFG_MARTINA.ipynb
| |-- fig_model_comparison.png
| |-- fig_ablation.png
| |-- fig_individual_differences.png
| |-- fig_correlaciones_individuales.png
| |-- fig_loso_barras.png
| |-- fig_performance_by_nrep.png
| |-- fig_permutation_importance.png
| |-- ml_results/
| | |-- model_performance.csv
| | |-- ml_model_results.csv
| | |-- ml_accuracy_per_subject.csv
| | |-- ml_accuracy_per_nrep.csv
| | |-- ml_permutation_importance.csv
| | |-- perm_importance_deci.csv
| | |-- perm_importance_switch.csv
| | |-- subject_accuracy_deci.csv
| | |-- subject_accuracy_switch.csv
|
|-- HSSM ANALYSES/
| |-- HSSM_TFG_MARTINA.ipynb

```

NOTE → subject EEG folders (DATA/preprocessament/EEG/s01/ to s27/) are not included in this GitHub repository because they could not be uploaded due to GitHub file size limitations.

Requirements

- Python >= 3.10
- mne
- numpy
- scipy
- pandas
- matplotlib

- seaborn
- scikit-learn
- jupyterlab

Additional packages used for the HSSM analyses :

- hssm==0.2.6
- pymc
- arviz
- blackjax

NOTE → the HSSM analyses were run in Google Colab. GPU acceleration is recommended because bayesian sampling with HSSM and PyMC can be computationally demanding.

Installation for the HSSM analyses :

```
pip install mne numpy scipy pandas matplotlib seaborn scikit-learn  
pip install hssm==0.2.6 pymc arviz blackjax
```